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TFPred: Learning discriminative representations from unlabeled data for few-label rotating machinery fault diagnosis

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ABSTRACT

Recent advances in intelligent rotating machinery fault diagnosis have been enabled by the availability of massive labeled training data. However, in practical industrial applications, it is often challenging and costly to annotate a large amount of data. To address the few-label fault diagnosis problem, a time–frequency prediction (TFPred) self-supervised learning framework is proposed to extract latent fault representations from unlabeled fault data. Specifically, the TFPred framework consists of a time encoder and a frequency encoder, with the frequency encoder to predict the low-dimensional representations of time domain signals generated by the time encoder with randomly augmented data. Subsequently, the pre-trained network is hooked with a classification head and fine-tuned with limited labeled data. Finally, the proposed framework is evaluated on a run-to-failure bearing dataset and a hardware-in-the-loop high-speed train simulation platform. The experiments demonstrate that the self-supervised learning framework TFPred achieved competitive performance with only 1% and 5% labeled data. Code is available at https://github.com/Xiaohan-Chen/TFPred.

1. Introduction

Rotating machinery is essential to modern manufacturing, but it is susceptible to failure due to prolonged exposure to harsh operating conditions. Factors such as high temperatures, high loads, and high rotation speeds can cause damage to the machinery, leading to costly repairs or even catastrophic failures (Chen, Yang, Zhong, Xi, & Liu, 2023; He, Wang, & Zhou, 2009; Lei, Yang, Jiang, Jia, Li, & Nandi, 2020; Li, Feng, et al., 2022; Zhou, Zhu, Shao, Dong, & Yang, 2023). Therefore, timely and accurate prediction of rotating machinery faults is critical to prevent these costly and potentially dangerous outcomes (Gao, He, Dong, Liu, & Lyu, 2022; He, Wang, Liu, Qin, & Zhou, 2017; He, Wang, Liu, & Zhou, 2013; Wang, Wang, Liu, Shen, & Dong, 2023; Yan, Shen, Sun, & Chen, 2019). Recent developments in sensor technology and the industrial internet of things have made it possible to collect and store large amounts of data about rotating machinery (Chen, Wang, Ding, Yi & Han, 2023; Hou, Wang, Dong, Hu, & Liu, 2022). This vast amount of data enables the training of intelligent fault diagnosis methods based on data-driven approaches, which have received significant attention from researchers and engineers in recent years (Chen, Zhang, & Gao, 2021; Liu, Liu, Jia, Zhang, & Tan, 2022; Lundgren & Jung, 2022; Shakiba, Shojaee, Azizi, & Zhou, 2022; Wen, Li, Hou, & Mu, 2022; Xu, Sheng, & Gao, 2021).

Supervised learning is a widely used approach for fault diagnosis, as it allows models to extract latent fault features from vibration signals without relying on prior knowledge (Chen, Yang, Xue, Huang, Ferrero, & Wang, 2023; Lu, Song, & Xu, 2021; Luo, Guan, Yang, & Smith, 2023; Wan, Yang, Huang, Zeng, & Liu, 2021). Notably, supervised learning methods require sensor measurements paired with explicit labels indicating the specific fault type present. However, annotating a large amount of high-quality fault labels is a costly and laborious process that requires human involvement, making it difficult in practical industrial applications. Additionally, annotating fault data from rotating machinery presents additional difficulties compared to tasks like computer vision or audio processing tasks, as it requires extensive expert knowledge and work experience. Consequently, acquiring a quantity of labeled data for model training can be challenging in rotating machinery. When the available labeled training data is limited, intelligent fault diagnosis models may encounter issues such as overfitting and the inability to identify discriminative features (Fang et al., 2022; Hu, Wu, Sun, Yan, & Chen, 2023; Ju, Tian, Liu, & Ma, 2021; Xue, Yang, Chen, Tian, & Wang, 2023).

The usage of semi-supervised learning-based fault diagnosis methods is popular in addressing the problem of insufficient labels by

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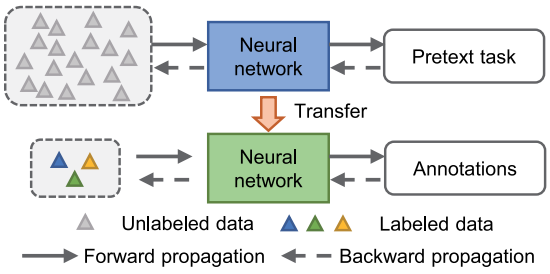


Fig. 1. An illustration of SSL for few-label data problems.

utilizing both labeled and unlabeled data. For example, Yu, Lin, Ma, Li, and Li (2021) applied the distribution discrepancies between the unlabeled data and the cluster center of labeled data to estimate pseudo-labels for unlabeled data and then introduced Kullback–Leibler divergence to reduce the distance between unlabeled data and cluster centers. Considering that the generated pseudo-label may be incorrect due to label errors and noise in practical applications, Chen, Yang, and Zhong (2021) proposed a graph-based semi-supervised learning model to annotate the unlabeled data, followed by a random forest algorithm to verify the prediction accuracy. In addition, Cheng, Liu, Zhou, and Yuan (2023) proposed an epoch-level and batch-level semi-supervised learning network to resolve noisy and erroneous label problems in practical industry. However, the performance of most semi-supervised learning-based fault diagnosis methods heavily relies on the accuracy of cluster centers and pseudo labels, bringing new challenges for rotating machinery fault diagnosis problems.

Deep transfer learning (DTL)-based method is also a popular method to solve data and label sparsity problem (Chen, Yang, Xue, Huang, et al., 2023; Yu, Yang, & Huang, 2022). DTL-based method reuses knowledge learned from source domain data and transfers it to target domains with limited labeled or unlabeled data. For example, Chen, Yang, Wen, and Guan (2021) proposed a domain adversarial network to learn inheritable features from the labeled source domain and unlabeled target domain via a zero-sum game. Zhao et al. (2021) conducted an extensive study on the DTL-based methods for rotating fault diagnosis problems and found that these methods effectively enhance predictive accuracy in scenarios with limited data and labels. Nevertheless, DTL-based methods still need a great number of annotated data, as they rely on the assumption that the source domain data are adequately labeled (Chen, Yang, Xue, Yang, Song, & Zhong, 2023).

Self-supervised learning (SSL) is a promising machine learning approach that allows models to extract latent representations without explicit labels. Models are given a set of unlabeled data with predefined pretext tasks to generate their own pseudo labels and learn features from the input data (Balestriero et al., 2023). Therefore, applying SSL to pre-train a feature extraction backbone with easily collected unlabeled data is an intuitive approach for addressing few-label problems. Subsequently, the fault diagnosis model can be initialized with the pre-trained backbone and fine-tuned with limited labeled data, as illustrated in Fig. 1. Among the various pretext tasks in SSL, instance discrimination is commonly used by training the model to differentiate between similar and dissimilar pairs of input data (Wu, Xiong, Yu, & Lin, 2018). Each data pair consists of a positive and a negative sample, where the positive sample is similar to the target data and the negative sample is dissimilar. The pretext task is designed to prompt networks to learn representations that effectively discriminate between similarity and dissimilarity.

SSL has led to significant improvements in computer vision tasks. For instance, to enlarge the dictionary size of negative samples and improve the model robustness, He, Fan, Wu, Xie, and Girshick (2020) proposed the momentum contrast (MoCo) strategy to dynamically update the key queue of data samples. Chen, Kornblith, Norouzi, and Hinton (2020) introduced a simple contrastive learning of representations (SimCLR) and emphasized the importance of combining multiple data augmentation methods for successful instance discrimination tasks. Grill et al. (2020) developed a Siamese network with two identical branches: online stream and target stream, where the online stream encodes the input data and predicts the target stream outputs, while the target stream encodes an augmented copy of the input image. With regard to rotating machinery fault diagnosis, Li, Huang, Chen, Xia, Chen, and Li (2022) applied a self-supervised cluster algorithm to address data imbalance problems. Considering the inter-instance and intra-temporal characteristics of time-series signals, Hu et al. (2023) integrated the instance discrimination and temporal relation SSL tasks for the semi-supervised fault classification. Building on the MoCo approach, Wang, Li, Li, Li, Chen, and Zhang (2023) incorporated meta-learning and the momentum encoder to propose a one-stage SSL method for open-set cross-domain fault diagnosis. Ding, Zhuang, Ding, and Jia (2022) applied MoCo to extract local representations from unlabeled run-to-failure vibration signals, effectively capturing incipient fault characteristics in rolling bearings. Based on contrastive predictive coding (Oord, Li, & Vinyals, 2018), Yang et al. (2023) constructed a pretext task that involves predicting future signal sequences to learn discriminative representations from unlabeled data. Akrim, Gogu, Vingerhoeds, and Salaün (2023) proposed an encoder–decoder self-supervised learning framework to learn underline features from time-series signals. Note that most of the SSL methods rely on constructing negative sample pairs to learn latent representations, since directly predicting the similarity of augmented samples may lead to model collapse (Grill et al., 2020). In addition, the existing methods still require a lot of labeled data to fine-tune the pre-trained model, typically exceeding 10% of the total labeled data (Hu et al., 2023).

To address the aforementioned challenges, as well as to reduce the reliance on labeled data and enhance the practical application of intelligent fault diagnosis methods, a time-frequency prediction (TFPred) self-supervised learning framework is proposed. The TFPred framework consists of a time encoder and a frequency encoder. The time encoder is responsible for generating temporal representations from time-series data, and the frequency encoder is responsible for generating frequency representations from frequency data. The frequency representations are then applied to predict the temporal representations via a prediction head. Intuitively, the frequency representations should be correlated with the corresponding temporal representations in the feature space. Therefore, the TFPred model is optimized to maximize the correlation between the frequency predictions and temporal representations to learn fault latent representations without labels. The main contributions of this paper are:

(1) The proposed novel SSL framework TFPred leverages both time and frequency domain data to learn latent representations from unlabeled data for rotating machinery fault diagnosis tasks.
(2) The cross-correlation matrix loss is introduced in this paper to measure the correlation between frequency domain predictions and time domain representations without relying on a large number of negative samples.
(3) A set of time domain data augmentation methods are specifically designed in this paper for improving the robustness and generalization of the model in rotating machinery fault diagnosis.
(4) The performance of the framework TFPred is evaluated on a run-to-failure bearing dataset and a hardware-in-the-loop high-speed train (HIL-HST) simulation platform, both of which have limited labeled fault data.

The rest of this paper is organized as follows. Section 2 describes the proposed TFPred framework in detail. Section 3 presents the case studies and results analysis of the TFPred. Section 4 concludes the paper and discusses future work.

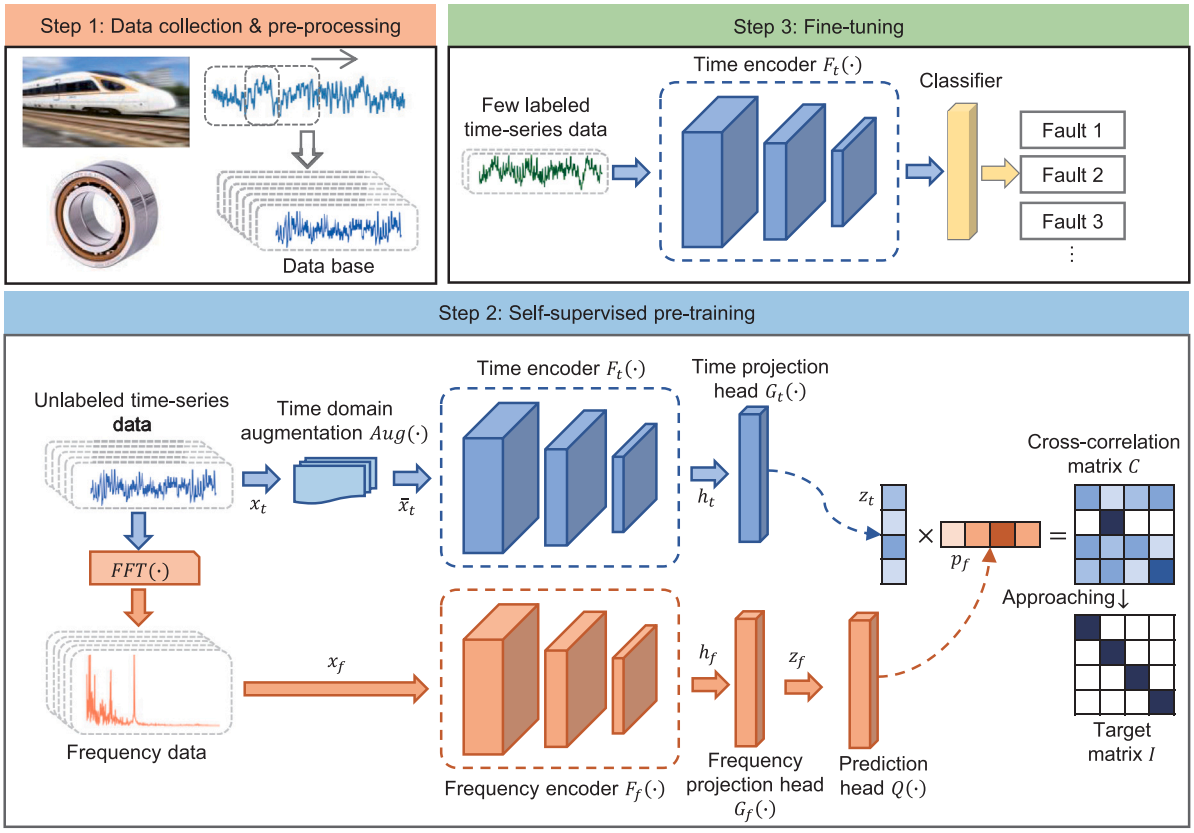


Fig. 2. A visualization of the proposed TFPred framework for few-label rotating machinery fault diagnosis.

2. Methodology

2.1. Overview

Annotating a large amount of fault data in rotating machinery fault diagnosis is costly and time-consuming. To effectively utilize the readily available unlabeled fault data when only a few labeled examples are available, a novel SSL framework, TFPred, is proposed in this paper to learn robust time and frequency representations for fault diagnosis tasks, as shown in Fig. 2. The overall framework is divided into three steps:

Step 1. Data collection and pre-processing: Acquire fault data from rotating machinery, split the time-series data into 1024-sample splits with an overlap of 512 data points, and scale them between 0 and 1.

Step 2. Self-supervised learning: Augment the time domain data and transform unlabeled time-series data into the frequency domain via fast Fourier transform (FFT). Then, simultaneously input both time and frequency domain data into the TFPred self-supervised learning framework to learn robust representations via a pre-defined pretext task. The cross-correlation loss function is specifically designed to optimize the model parameters.

Step 3. Fine-tuning: Retain the pre-trained time encoder after *Step 2* and connect it to a classifier head, then fine-tune the learned representations with limited labeled data.

Remark. It is worth noting that the frequency encoder may also be retained and transferred to downstream tasks, since both the time and frequency encoders learn latent representations in their respective domains. However, to ensure ease of implementation and fair comparison, only the time encoder is retained and transferred to *Step 3*.

2.2. Data augmentation

The SSL benefits greatly from employing strong data augmentation methods to learn latent representations from raw signals (Chen et al., 2020). In addition, data augmentation plays an important role in generating different views of positive and negative sample pairs, which allows the model to learn robust representations. Although positive and negative sample pairs are not utilized in this study, the experiments show that appropriate data augmentations can still improve the model's ability to learn from a limited amount of data. Therefore, a set of time-domain data augmentation methods are specifically designed in this paper.

Data augmentation methods are well-established in computer vision, but they are less common in vibration signal analysis. Wang, Li, et al. (2023) proposed four vibration signal augmentation methods: horizontal flip, vertical flip, center flip and Gaussian noise. To expand the training data and improve the model's robustness, this paper further introduces four additional time-domain signal augmentation methods: mask, gain, shift, and fade.

Gaussian SNR noise: In the study by Wang, Li, et al. (2023), random Gaussian noise with a mean of 0 and a standard deviation of 0.01 is added to the original signal to simulate real-world noise commonly encountered in signal processing applications. However, this approach does not consider the different levels of noise observed in practical scenarios. To address this limitation, Gaussian SNR (signal-to-noise ratio) noise is introduced in this paper to simulate different noise levels. During each training epoch, noise signals are generated by randomly selecting the SNR from a uniform distribution ranging from 3 to 30 dB.

Mask: A mask is applied to randomly set a consecutive sequence of t_{mask} time steps $[t_0, t_0 + t_{\text{mask}}]$ to zero in the signal, where t_{mask} is randomly selected from a uniform distribution between 100 to 450. Similarly, the starting point of the mask t_0 is also randomly selected,

Table 1
Random vibration signal augmentations.

Augmentation	Probability	Description
Gaussian SNR noise	80%	Inject Gaussian noise into the vibration signal with randomly selected SNR
Mask	50%	Mask a part of vibration signal with randomly selected length
Gain	30%	Randomly increase or decrease the vibration signal value
Shift	30%	Shift the vibration signal forward or backward with randomly selected steps
Fade	20%	Gradually increase or decrease the vibration signal value
Horizontal flip	20%	Horizontally flip the vibration signal
Vertical flip	20%	Vertically flip the vibration signal

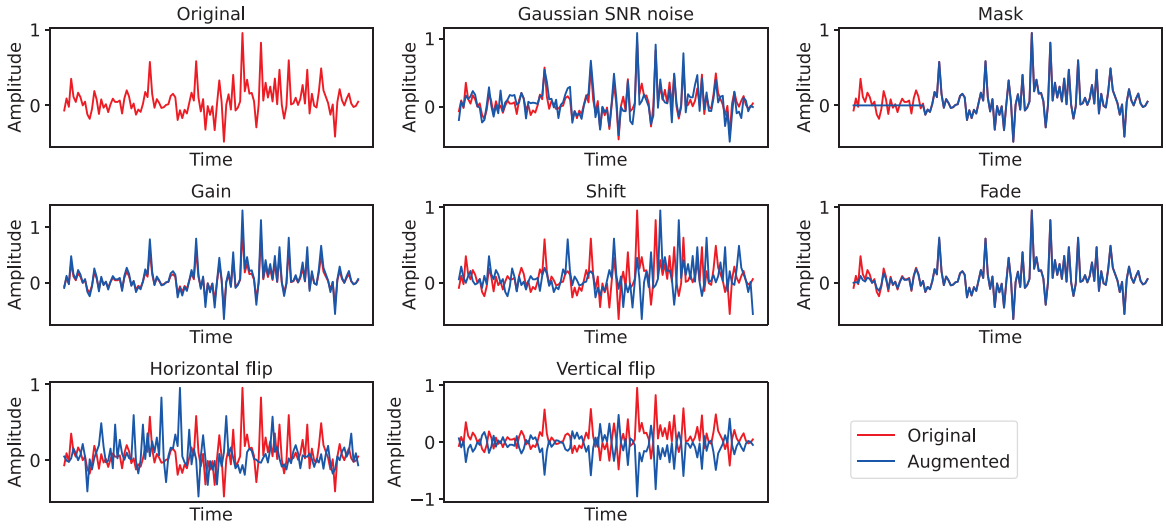


Fig. 3. Visualizations of the seven augmentation methods on bearing fault signals.

with restriction that the masked region does not exceed the signal length of 1024, i.e., $t_0 + t_{\text{mask}} < 1024$.

Gain: The vibration signal value can be adjusted by multiplying a gain factor τ, where $\tau \in [0.5, 1.5]$. A gain factor of $\tau < 1$ will decrease the vibration value, while a gain factor of $\tau > 1$ will increase the vibration value. The gain factor is randomly selected from a uniform distribution within the specified range.

Shift: The time axis of a vibration signal can be shifted forward or backward by a randomly selected number of steps $t_{\text{shift}} \in [-300, 300]$. A negative shift value means a backward shift, while a positive shift value means a forward shift.

Fade: The amplitude of a vibration signal can be logarithmically increased or decreased over a randomly selected number of signal points, uniformly distributed between 0 and 300.

Horizontal flip: The time axis of a vibration signal can be flipped by reversing the order of the signal's samples.

Vertical flip: The amplitude axis of a vibration signal can be flipped by reversing the sign of the signal's samples.

Fig. 3 illustrates the effects of the signal augmentation methods mentioned above. Within each training batch, the seven augmentation methods are sequentially applied to time-series data based on predefined probability, as shown in Table 1. Importantly, this process maintains the original batch size, despite generating diverse versions of the time-series data, as shown in Fig. 4. The proposed augmentation methods can create a diverse dataset to train the self-supervised learning model and improve the model's robustness to noise. Notably, these data augmentation techniques are not employed during the subsequent fine-tuning stage with labeled data, simplifying the implementation process for practical applications. This two-stage approach strikes a balance between enriching the model's feature representation and maintaining a straightforward application for real-world deployments.

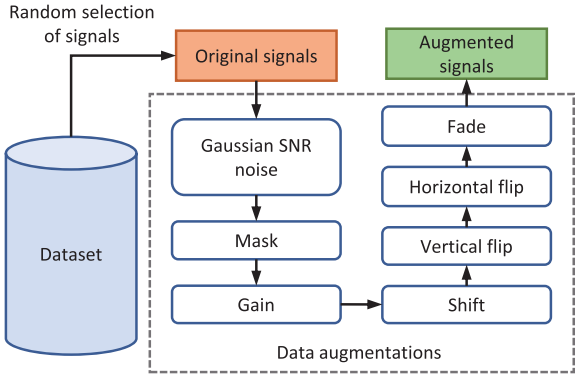


Fig. 4. An illustration of the data augmentation process during training.

2.3. Self-supervised learning

A novel SSL framework is proposed in this paper that applies the frequency domain latent representations to predict augmented time domain representations. Intuitively, the frequency domain features should be correlated with its corresponding augmented time domain representations, but uncorrelated with the time domain representations of the remaining samples.

The TFPred framework consists of various components, including a time encoder $F_t(\cdot)$, a frequency encoder $F_f(\cdot)$, a time projection head $G_t(\cdot)$, a frequency projection head $G_f(\cdot)$, and a prediction head $Q(\cdot)$, as shown in Fig. 2. The time and frequency encoders learn two representations from time and frequency domain data respectively. Specifically, the time representation captures the temporal dynamics of the data, while the frequency representation captures the spectral content. These two representations are then mapped into low-dimensional spaces via

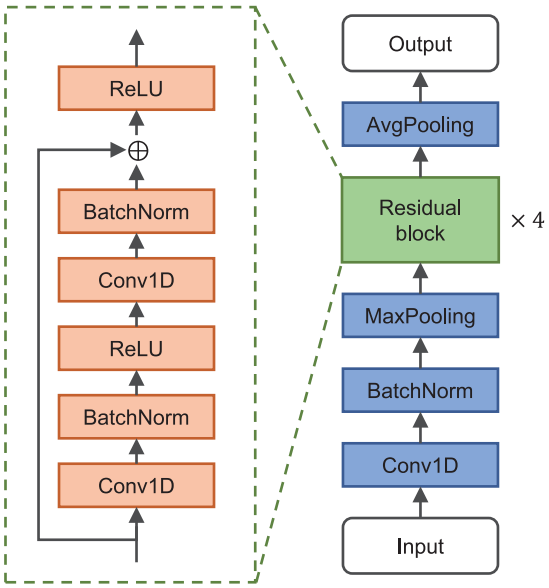


Fig. 5. The structure of ResNet-18: Conv1D represents 1D convolution, BatchNorm represents batch normalization, MaxPooling and AvgPooling represent max pooling and average pooling respectively, ReLU represents rectified linear unit activation function, and Residual block ×4 means that the residual block is stacked four times.

the time and frequency projection heads respectively. Finally, a prediction head is applied to predict the time domain representation corresponding to the frequency representation.

Time and frequency encoders: The time and frequency encoders apply the 1D ResNet-18 network (He, Zhang, Ren, & Sun, 2016) (without classification layer) shown in Fig. 5 to extract representations in this paper. The first three layers of ResNet-18 follow the structure of the conventional convolutional neural network (CNN), while the subsequent layers utilize four residual blocks. A residual block is a building block of ResNet that consists of two convolutional layers, followed by a shortcut connection which adds the output of the first convolutional layer to the output of the second convolutional layer. The time and frequency encoders share the same network architecture in this paper, but they do not share parameters. Consequently, the time and frequency encoder parameters are independently updated through gradient backpropagation.

To begin, the time-series data x_t undergo random augmentation to obtain augmented data $\bar{x}_t = \mathrm{Aug}(x_t)$, where $\mathrm{Aug}(\cdot)$ represents the augmentation function. Simultaneously, the FFT is applied to transform the time domain data x_t to the frequency domain representation $x_f = \mathrm{FFT}(x_t)$. Next, the augmented time domain data $\bar{x}_t$ and the corresponding frequency domain data x_f are simultaneously input into the time and frequency encoders to obtain time-domain representation $h_t = F_t(\bar{x}_t)$ and frequency-domain representation $h_f = F_f(x_f)$, respectively.

Time and frequency projection heads: Linear time and frequency projection heads are applied to project the extracted representations into a low-dimensional subspace of 128 dimensions, resulting in two 128-dimensional feature vectors: $z_t = G_t(h_t)$ representing the time features, and $z_f = G_f(h_f)$ representing the frequency features.

Prediction head: To predict the corresponding augmented time features z_t using the frequency features z_f, a nonlinear prediction head $Q(\cdot)$ is employed. The prediction head consists of two linear layers with 256 and 128 hidden units respectively, with the first layer followed by a batch normalization layer and a ReLU activation function. Table 2 comprehensively presents the dimensions of each feature representation encountered throughout the network architecture.

Loss: Cross-correlation loss (Zbontar, Jing, Misra, LeCun, & Deny, 2021) is introduced in this paper to measure the correlations between frequency predictions $p_f = Q(z_f)$ and augmented time features z_t. The

Table 2
Feature representation size dynamics in TFPred.

Module/Transformation	Input size	Output size
Time augmentation $Aug(\cdot)$	x_t: [B, 1024]	$\bar{x}_t$: [B, 1024]
Time encoder $F_t(\cdot)$	$\bar{x}_t$: [B, 1024]	h_t: [B, 512]
Time projection head $G_t(\cdot)$	h_t: [B, 512]	z_t: [B, 128]
FFT	x_t: [B, 1024]	x_f: [B, 512]
Frequency encoder $F_f(\cdot)$	x_f: [B, 512]	h_f: [B, 512]
Frequency projection head $G_f(\cdot)$	h_f: [B, 512]	z_f: [B, 128]
Prediction head $Q(\cdot)$	z_f: [B, 128]	p_f: [B, 128]

B represents batch size.

underlying idea is that the frequency prediction for a given sample should be correlated with its corresponding augmented time feature, while being uncorrelated with the time features of the remaining samples in the batch dimension. Thus, the cross-correlation matrix between frequency predictions p_f and augmented time features z_t should be close to the identity matrix. In other words, the diagonal elements should approach 1, whereas the off-diagonal elements should approach 0. This is the definition of the loss function:

$$\mathcal{L} = \sum_{i=1}^{N}(1 - C_{ii})^2 + \lambda \sum_{i=1}^{N}\sum_{j=1, j\neq i}^{N} C_{ij}^2 \tag{1}$$

where λ is a constant trading off the contribution of the uncorrelation term, N is the dimension of p_f and z_t, i and j are the index of dimension, and C is the cross-correlation matrix between the p_f and z_t along the batch dimension:

$$C = \frac{1}{N}\frac{p_f^T}{\|p_f\|_2}\frac{z_t}{\|z_t\|_2} \tag{2}$$

where $\|\cdot\|_2$ is the l_2-norm, ensuring p_f and z_t reside within a comparable range, and C's convergence towards an N-by-N identity matrix.

The cross-correlation loss function employed in this paper is similar to the widely used SSL loss function InfoNCE (Chen et al., 2020). However, unlike InfoNCE, the cross-correlation loss function does not need to construct a significant quantity of negative samples, making models efficient and scalable. By employing the cross-correlation loss function, the model learns latent representations that discriminate time and frequency correlations. The algorithm for TFPred is outlined in pseudocode in Algorithm 1.

3. Experiments

3.1. Datasets

3.1.1. Run-to-failure bearing dataset

The run-to-failure bearing dataset was created by simulating realistic bearing damages through accelerated lifetime tests (Lessmeier, Kimotho, Zimmer, & Sextro, 2016), as shown in Fig. 6. After the accelerated lifetime tests, except health bearings, pitting (fatigue damage) and indentation (plastic deformation) faults are observed on the outer ring (OR), inner ring (IR), or both OR and IR of bearings, as shown in Fig. 7. The extents of the damage are divided into three categories: minor (damage length $\leq$ 2 mm), medium (damage length $>$ 2 and $\leq$ 4.5 mm), and serious (damage length $>$ 4.5 mm). Detailed fault information and corresponding labels are provided in Table 3. Motor current signals are measured by current transducers at a sampling rate of 64 kHz. The experiments are conducted under four operating conditions: (1) rotating speed 1500 rpm, load torque 0.7 Nm, and radial force 1000 N (code WC1); (2) rotating speed 900 rpm, load torque 0.7 Nm, and radial force 1000 N (code WC2); (3) rotating speed 1500 rpm, load torque 0.1 Nm, and radial force 1000 N (code WC3); (4) rotating speed 1500 rpm, load torque 0.7 Nm, and radial force 400 N (code WC4). The run-to-failure bearing dataset contains 3000 instances, with 1024 sample points per instance. The data are split into training (70%), testing (20%), and validation (10%) sets. Only 1% of the training data is labeled, while the remaining data is unlabeled.

Algorithm 1 TFPred SSL algorithm.

Input: Time augmentation function Aug, fast Fourier transform function FFT, time encoder $F_t(\cdot)$, frequency encoder $F_f(\cdot)$, time projection head $G_t(\cdot)$, frequency projection head $G_f(\cdot)$, prediction head $Q(\cdot)$, training epoch V, epoch index v, batch size B, batch index b, dataset $\mathcal{D}$

Output: Time encoder $F_t(\cdot)$

Randomly initialize $F_t(\cdot)$, $F_f(\cdot)$, $G_t(\cdot)$, $G_f(\cdot)$, and $Q(\cdot)$
for $v \in \{1, \dots, V\}$ **do**
  for $b \in \{1, \dots, B\}$ **do**
    Sample time domain data from dataset $x_t \sim \mathcal{D}$
    Augment time domain data:

$$\tilde{x}_t \leftarrow \mathrm{Aug}(x_t)$$

    Obtain frequency domain data:

$$x_f \leftarrow \mathrm{FFT}(x_t)$$

    Obtain time and frequency representations:

$$h_t \leftarrow F_t(\tilde{x}_t)$$

$$h_f \leftarrow F_f(x_f)$$

    Transform representations to low dimension:

$$z_t \leftarrow G_t(h_t)$$

$$z_f \leftarrow G_f(h_f)$$

    Predict time features:

$$p_f \leftarrow Q(z_f)$$

    Compute cross-correlation matrix:

$$C \leftarrow \frac{1}{N} \frac{p_f^T}{\|p_f\|_2} \frac{z_t}{\|z_t\|_2}$$

    Compute loss:

$$\mathcal{L} \leftarrow \sum_{i=1}^{N}(1 - C_{ii})^2 + \lambda \sum_{i=1}^{N} \sum_{j=1, j \neq i}^{N} C_{ij}^2$$

    Update $F_t(\cdot)$, $F_f(\cdot)$, $G_t(\cdot)$, $G_f(\cdot)$, and $Q(\cdot)$
  end for
end for

Table 3
Fault descriptions of run-to-failure bearing dataset.

Label	Fault	Location	Damage length	Mode
0	Normal	–	–	–
1	Indentation	OR	≤ 2 mm	S
2	Pitting	OR	> 2 mm	S
3	Pitting	OR	≤ 2 mm	S
4	Indentation	OR	≤ 2 mm	M
5	Pitting	IR+OR	> 2 mm	S
6	Pitting	IR+OR	> 4.5 mm	M
7	Indentation	IR+OR	≤ 2 mm	M
8	Pitting	IR	≤ 2 mm	S
9	Pitting	IR	> 4.5 mm	S

S: single point damage, M: multiple point damage.

3.1.2. HIL-HST simulation platform

The HIL-HST simulation platform (Chen, Chen, Fan, Peng, & Yang, 2022; Yang, Yang, Peng, Chen, Liu, & Gui, 2018) is illustrated in Fig. 8. The platform consists of three types of faults: broken rotor bar (BRB), stator inter-turn short-circuits (SIS), and air gap eccentricity (AGE). Detailed fault information and labels are presented in Table 4.

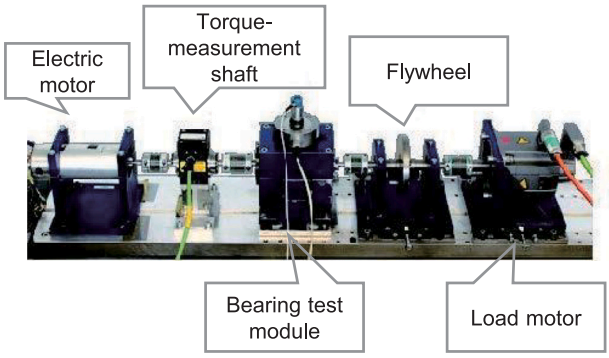


Fig. 6. The testbench of the PU dataset.

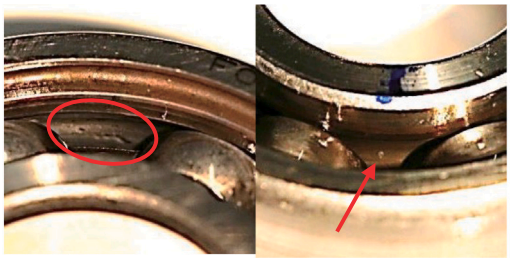


Fig. 7. OR indentation fault (left) & IR pitting fault (right).

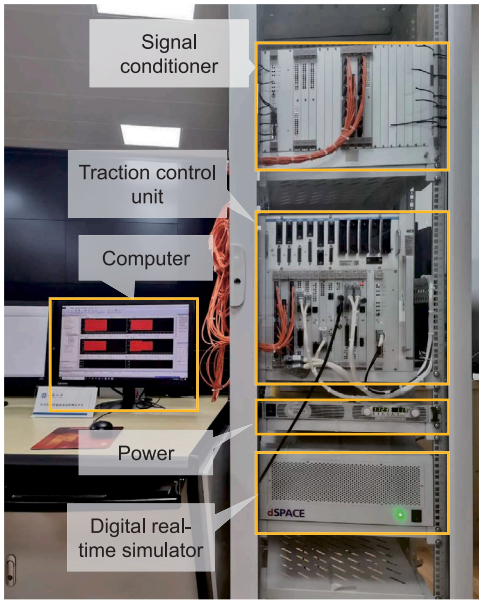


Fig. 8. The HIL-HST simulation platform.

Faults are generated using computer simulations and injected into the traction motor via a digital real-time digital simulator. Each fault can be simulated with three different fault extents: minor, medium, and serious. Fault data are collected at three speeds: 20, 169, and 280 km/h, represented by codes WC1, WC2, and WC3, respectively. The HIL-HST simulation platform contains 800 instances, with 1024 sample points per instance. The data split follows the same distribution as the run-to-failure bearing dataset, with only 5% of the training data being labeled, while the remaining data is unlabeled. Table 5 details the precise number of data allocated to each split for each dataset.

Table 4
Fault descriptions of HIL-HST simulation platform.

Label	Fault	Level
0	Normal	–
1	BRB	Minor
2	BRB	Medium
3	BRB	Serious
4	SIS	Minor
5	SIS	Medium
6	SIS	Serious
7	AGE	Minor
8	AGE	Medium
9	AGE	Serious

Table 5
The number of data split for training, validation, and testing sets.

Dataset	Number of data			
	Training		Validation	Test
	Labeled	Unlabeled		
Run-to-failure bearing dataset	30	2070	300	600
HIL-HST simulation platform	30	530	160	80

3.2. Compared methods

The proposed method, TFPred, is compared with a conventional supervised learning method and six state-of-the-art semi-supervised and SSL methods:

(1) Supervised: only applies labeled data to train the model to predict faults in a supervised manner.
(2) Mean teacher (Tarvainen & Valpola, 2017): leverages unlabeled data to improve the performance with limited labeled data through a teacher-student paradigm. The student predicts labels, while the teacher guides the student towards consistent predictions with teacher outputs.
(3) MoCo (He et al., 2020): constructs a large and consistent dictionary of representations through momentum-updated encoders, fostering robust self-supervised learning.
(4) SimCLR (Chen et al., 2020): maximizes the agreement between representations of augmented views of the same input, using a contrastive loss. Moreover, a two-layer nonlinear projection head is proposed to improve the learned representations.
(5) BYOL (Grill et al., 2020): employs a two-stream learning process. An online network predicts the target network representation of the same input with different augmentations. The target network parameters are updated with a moving average of the online network.
(6) SimSiam (Chen & He, 2021): utilizes a Siamese network architecture with two identical branches processing augmented views of the same input. These branches predict the representations of each other, promoting the learning of consistent representations.
(7) Barlow twins (Zbontar et al., 2021): focuses on minimizing redundancy in representations of augmented views by predicting the residual between representations of an augmented view and its noisy counterpart.

To ensure that all methods are compared fairly, all methods in this study, including the proposed approach and its baselines, employed the same 1D ResNet-18 encoder to extract latent representations. The model is trained using stochastic gradient descent with an empirical momentum hyperparameter of 0.9 for 500 epochs with a batch size of 256. The initial learning rate is 0.01 using a cosine annealing schedule (Loshchilov & Hutter, 2016). The trade-off parameter λ of loss function (1) is set to 0.005, following the setting provided in Zbontar et al. (2021).

Table 6
An illustration of the four compared augmentation sets and the corresponding codes.

Code	Augmentation methods
Aug1	Horizontal flip, vertical flip, center flip, Gaussian noise
Aug2	Horizontal flip, vertical flip, center flip, Gaussian SNR noise
Aug3	Horizontal flip, vertical flip, center flip, Gaussian SNR noise, mask, gain, shift, fade
Aug4	Horizontal flip, vertical flip, Gaussian SNR noise, mask, gain, shift, fade

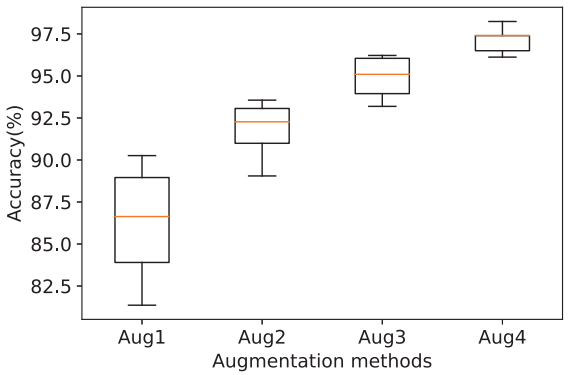


Fig. 9. Augmentation methods ablation experiments.

3.3. Ablation study

Wang, Li, et al. (2023) proposed several augmentations for time-series signal fault diagnosis, including horizontal, vertical, center flip, and Gaussian noise (code Aug1). However, the noise distribution in their proposed augmentation methods is constant and may not simulate the complex noise in real environments. To address this limitation, Gaussian noise is replaced with Gaussian SNR noise to simulate different SNR levels in practical applications (code Aug2). Furthermore, four additional data augmentation methods – mask, gain, shift, and fade – are proposed to further expand the training data and enhance model robustness (code Aug3). It is worth noting that the experiments show that center flip can decrease the model performance and hinder the model from learning discriminative representations from raw signals. Therefore, the center flip augmentation method is removed from the augmentation bank in this paper (code Aug4). The above-mentioned data augmentation methods are listed in Table 6.

An ablation study is conducted to evaluate the influences of the four data augmentation methods on WC3 of the run-to-failure bearing dataset for SSL. As shown in Fig. 9, the average accuracy increases significantly when Gaussian noise is replaced with Gaussian SNR noise (Aug2). This finding suggests that the model generalizes well to different noise levels when trained with various SNR noises. In addition, augmentations with mask, gain, shift and fade (Aug3) also significantly increase the model's accuracy (Aug3). These augmentation methods help the model learn robust representations by introducing variations in the amplitude, displacement, and intensity of the signal. It is evident from Aug4 that removing the center flip from the augmentation bank improves average accuracy by approximately 1%. This may be due to the center flip creating mirror images of the signal and destroying the temporal information of signals. Consequently, the model may mistake the augmented signal features as real features, resulting in decreased accuracy.

3.4. Case study 1: Run-to-failure bearing fault

The fault diagnosis accuracy and standard deviation of the eight methods, using 1% labeled data under different working conditions, are presented in Table 7. Overall, TFPred, the proposed self-supervised

Table 7
Fault diagnosis accuracy (%) and standard deviation under different working conditions with 1% labeled data in run-to-failure bearing dataset. **Bold** and underline represents the best and second best results respectively.

Working condition	WC1	WC2	WC3	WC4
Supervised	61.30 ± 12.57	59.52 ± 10.63	60.90 ± 14.62	71.82 ± 5.81
Mean teacher	76.53 ± 3.27	68.73 ± 3.03	77.94 ± 2.71	82.74 ± 1.20
MoCo	86.88 ± 3.42	91.50 ± 1.66	87.85 ± 3.01	93.10 ± 2.34
SimCLR	96.87 ± 0.94	92.24 ± 0.90	93.18 ± 0.80	95.79 ± 1.52
BYOL	78.93 ± 1.47	67.81 ± 4.50	68.58 ± 0.76	73.19 ± 4.12
SimSiam	90.10 ± 3.22	85.03 ± 3.98	77.09 ± 5.49	85.00 ± 3.10
Barlow twins	93.86 ± 1.89	80.20 ± 6.90	85.67 ± 3.51	90.80 ± 2.51
TFPred	**98.17 ± 0.59**	**96.20 ± 1.87**	**97.13 ± 0.65**	**97.75 ± 0.42**

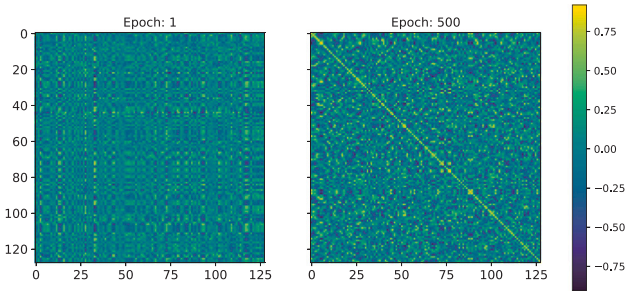


Fig. 10. Visualization of cross-correlation C.

Table 8
Fault diagnosis accuracy (%) and standard deviation under different working conditions with 5% labeled data in HIL-HST simulation platform. **Bold** and underline represents the best and second best results respectively.

Working condition	WC1	WC2	WC3
Supervised	49.50 ± 3.59	66.80 ± 3.45	57.30 ± 5.60
Mean teacher	61.18 ± 5.04	74.92 ± 4.32	76.64 ± 2.19
MoCo	82.84 ± 4.60	85.42 ± 5.15	85.58 ± 2.49
SimCLR	84.25 ± 4.34	89.58 ± 4.58	85.96 ± 0.96
BYOL	84.04 ± 3.93	90.83 ± 1.17	86.67 ± 1.95
SimSiam	84.38 ± 4.37	90.30 ± 1.38	83.81 ± 6.42
Barlow twins	75.00 ± 4.32	83.20 ± 3.78	71.90 ± 2.34
TFPred	**84.86 ± 1.93**	**91.84 ± 1.04**	**87.19 ± 1.43**

learning method, performs well in all working conditions, demonstrating its promising potential for few-labeled fault diagnosis problems.

TFPred achieves the highest average accuracy in all working conditions, surpassing the second-best method, SimCLR, by at least 1%. Moreover, the standard deviation of TFPred is also lower than that of SimCLR in three out of four working conditions. While MoCo, BYOL, SimSiam, and Barlow twins also achieve competitive results, their average accuracy is lower and their standard deviation is higher compared to TFPred. The optimal performance of TFPred can be attributed to two reasons. Firstly, the limited batch size hinders the ability of other SSL methods to extract discriminative features from insufficient negative samples. Secondly, these methods solely focus on time-series data but neglect frequency data. Instead, TFPred does not rely on a large batch size of negative samples to learn latent features from raw data, and it takes into account the relationship between time and frequency data. Both reasons enable TFPred to achieve higher and more robust performance than other methods. The supervised learning model cannot learn discriminative features from raw data when the labeled training data are limited, resulting in an average accuracy of only 63.39% across the four working conditions.

Fig. 10 visually depicts the assumption: the frequency prediction for a given sample should be correlated with its corresponding augmented time feature, while being uncorrelated with the time features of the remaining samples. In the initial stages of training (epoch 1), the cross-correlation matrix C reflects the limited learning progress of the time and frequency encoders, preventing them from accurately capturing the desired relationships between predicted frequencies and their corresponding time features. However, after extensive training (epoch 500), a clear pattern emerges: the diagonal elements of C exhibit significantly higher correlation scores compared to the non-diagonal elements. This observation validates the assumption and demonstrates the model's successful learning of the inherent relationships between frequency predictions and associated time features.

To demonstrate the latent representation learning ability of the eight models, the feature maps of the models after fine-tuning using WC3 test data are visualized using t-distributed stochastic neighbor embedding (t-SNE) dimensionality reduction techniques, as shown in Fig. 11. In terms of supervised learning, it is clear that all classes except class 6 are mixed together. This illustrates why supervised learning methods have relatively low accuracy when labeled data are limited. Specifically, SimCLR, SimSiam, and Barlow twins can effectively learn discriminative latent representations of class 2, 6, and 9, but the remaining classes are mixed together. Additionally, it is worth noting that MoCo, SimCLR, BYOL, SimSiam, and Barlow twins cannot well-separate class 1 and 7, because both classes are minor indentation faults with different fault locations and characteristics. The fault features of the two classes are similar, making it challenging for the models to learn discriminative representations. However, in comparison to the other methods, TFPred successfully clusters the latent representations of the ten classes in low-dimensional space, even when the labeled data are limited. This suggests that TFPred is able to learn discriminative features from a significant amount of unlabeled data. As a result, only a small amount of labeled data is needed to fine-tune the pre-trained model to achieve competitive results.

3.5. Case study 2: HIL-HST simulation platform

In order to further test the performance of the proposed TFPred model, eight methods are tested on the HIL-HST simulation platform under different working conditions with 5% labeled data. The experimental results are shown in Table 8. Supervised learning achieves the lowest accuracy among the three working conditions, with an average accuracy of 57.87%, and a relatively high standard deviation. This suggests that supervised learning is ineffective for fault diagnosis when labeled data is extremely limited. Although mean teacher method improves the average accuracy, it was not as high as the other SSL methods. MoCo, SimCLR, BYOL, SimSiam, and Barlow twins achieve competitive results in the three working conditions, with SimSiam's average accuracy being very close to that of TFPred, especially in WC1. TFPred still achieves the best accuracy in all three operating conditions, with a relatively low standard deviation. This suggests that TFPred is an effective SSL method for fault diagnosis when labeled data are limited.

Fig. 12 presents the confusion matrices of TFPred on the simulation platform. The model performs variably under different working conditions, with overall prediction accuracy acceptable. Note that the TFPred model achieves 100% accuracy for the SIS and AGE severe faults in all three working conditions. The prediction accuracy on BRB severe faults exceeds 87%, suggesting that TFPred is able to learn discriminative features from unlabeled data for severe faults. In addition, the TFPred

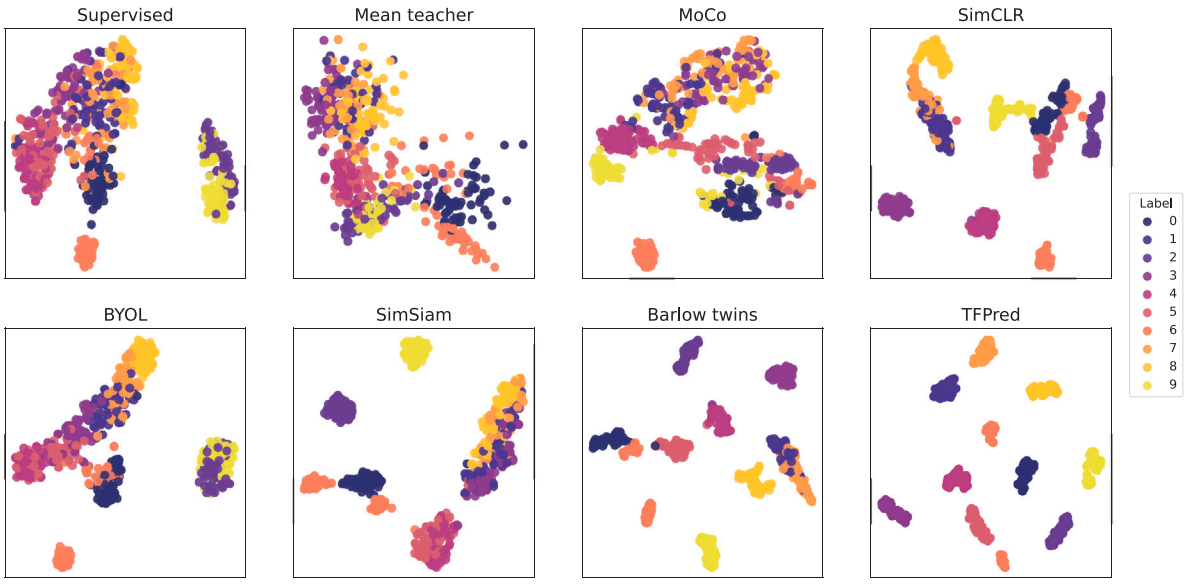


Fig. 11. Visualization of features of the eight methods on run-to-failure bearing dataset in WC2.

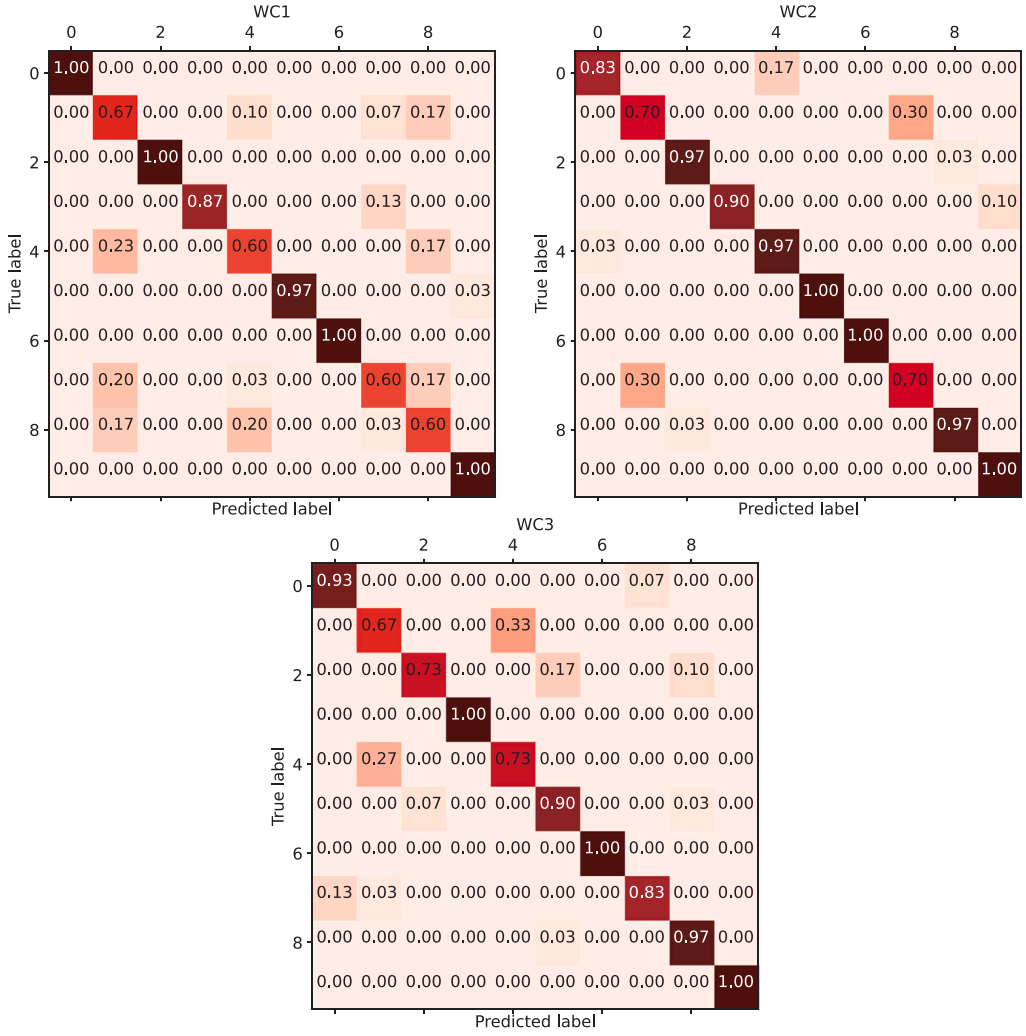


Fig. 12. Confusion matrix of TFPred model on HIL-HST simulation platform.

model also achieves average accuracies of 90%, 95%, and 85% for BRB, SIS, and AGE medium faults, respectively. Overall, TFPred is a promising SSL method for learning latent features from a large number of unlabeled data. The pre-trained TFPred model can be applied to

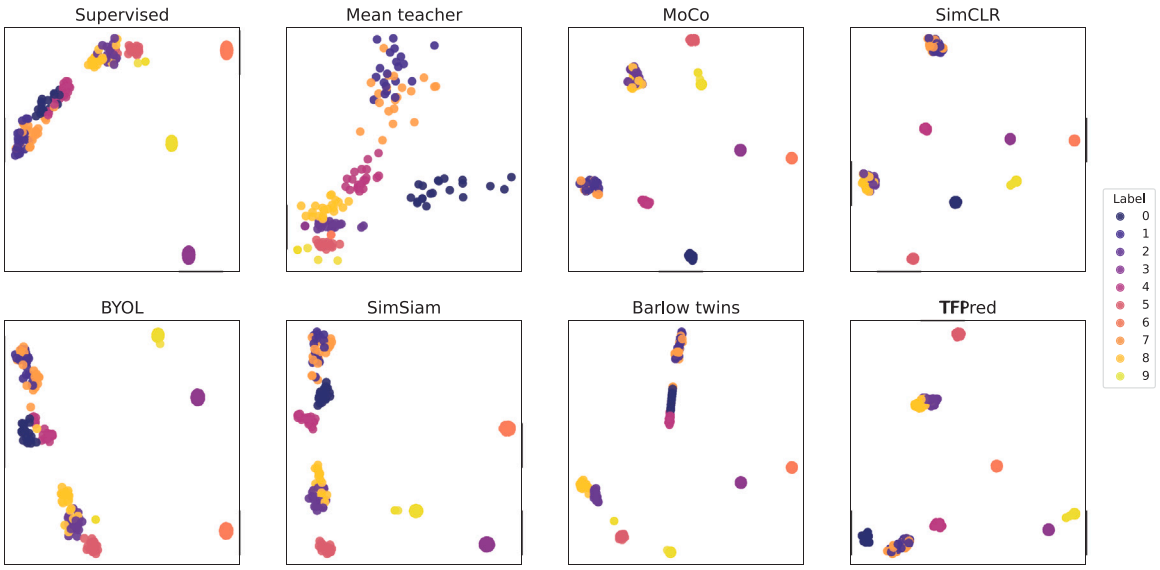


Fig. 13. Visualization of features of the eight methods on HIL-HST simulation platform in WC2.

predict rotating machinery faults in practical industry applications when only limited labels are available.

Fig. 13 visualizes the latent space representation of the ten fault categories after fine-tuning in WC2 using t-SNE dimensionality reduction. Notably, for supervised learning methods (excluding class 3, 6, and 9), the categories exhibit significant overlap, indicating the limitations of learning effective discriminative features from limited labeled data. MoCo, SimCLR, BYOL, SimSiam, and TFPred successfully separate class 3, 6, and 9 from the others, aligning with the 100% prediction accuracy for these categories achieved by the proposed method in the confusion matrix. However, these methods struggle to distinguish class 2 and 8. Notably, TFPred exhibits comparatively distinct classification boundaries and clustering centers for these classes. In conclusion, TFPred demonstrates superior feature learning capabilities compared to existing state-of-the-art methods, as evidenced by its superior performance in both visualization and quantitative evaluations.

4. Conclusion

In this paper, a novel SSL framework, TFPred, is proposed to resolve the problem of limited labeled data for rotating machinery fault diagnosis in practical industry. Specifically, TFPred consists of a time encoder and a frequency encoder. Time-series data are augmented and transformed into the frequency domain, and then input into the time encoder and frequency encoder, respectively. The frequency representations are subsequently employed to predict the time representations after augmentation. Finally, the pre-trained encoder is transferred to the downstream few-label fault diagnosis tasks. Moreover, a set of time-domain data augmentation methods are proposed in this paper, which significantly enhances the model's performance. The TFPred model is evaluated on a run-to-failure bearing dataset and a HIL-HST simulation platform, with only 1% and 5% labeled data used for fine-tuning the pre-trained model. The experiments suggest that TFPred is a promising method for few-labeled fault diagnosis problems. Although TFPred achieves competitive results compared to the state-of-the-art methods, experimental analysis shows that TFPred faces challenges in learning discriminative representations of minor faults. Therefore, future research will explore the model's performance on minor faults by incorporating additional data augmentation or feature engineering methods.

CRediT authorship contribution statement

Xiaohan Chen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Rui Yang:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Yihao Xue:** Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Baoye Song:** Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing. **Zidong Wang:** Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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